

# A Novel Solubilizing Fully $\pi$ -Conjugated Ladder Polymer and Its Thermoelectric Properties

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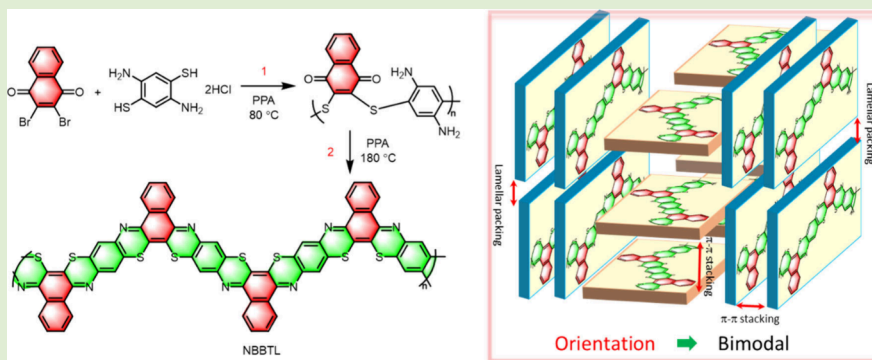
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**ABSTRACT:** Fully conjugated ladder polymers (CLPs) have emerged as promising candidates for next-generation organic electronics due to their extended  $\pi$ -conjugation pathways, intrinsic charge transport anisotropy, and exceptional chemical and thermal stability. However, insolubility remains a critical bottleneck, primarily stemming from side reactions that disrupt the linear structure. Herein, we report a soluble CLP, benzo[a]benzo[5,6][1,4]thiazino[3,2-c]phenothiazine ladder (NBBTL), via a stepwise strategy to preserve the linear structure of the polymer. The resulting polymer exhibits unprecedented solution processability while maintaining structural integrity, enabling the first reported thermoelectric generator fabrication. Thermoelectric performance evaluation revealed that  $\text{FeCl}_3$ -doped NBBTL achieved a conductivity of  $3.18 \pm 0.33 \text{ S cm}^{-1}$  and a power factor of  $0.94 \pm 0.09 \mu\text{W m}^{-1} \text{ K}^{-2}$ . Further analysis of the polymer chain orientation in films by grazing-incidence wide-angle X-ray scattering revealed that NBBTL exhibits a bimodal orientation with both face-on and edge-on. This study demonstrates that strategic molecular engineering can overcome long-standing insolubility issues in CLPs, potentially paving the way for their wider application in organic electronics and thermoelectric energy conversion.

Fully conjugated ladder polymers (CLPs) characterized by double-stranded fused-ring structure represent a promising class of interesting organic electronic materials due to their extended conjugation pathway along a rigid ladder-like backbone.<sup>1–6</sup> This unique double-stranded fused-ring structure imparts significant rigidity to the polymer backbone, effectively suppressing molecular chain motion. These features not only ensure exceptional thermal and chemical stability but also facilitate highly ordered molecular packing, thereby enhancing charge transport efficiency.<sup>7–9</sup> These exceptional properties make CLPs ideal for advanced electronic applications, including organic light-emitting diodes (OLEDs),<sup>10,11</sup> organic field-effect transistors (OFETs),<sup>12–15</sup> organic electrochemical transistors (OECTs),<sup>16–19</sup> organic thermoelectrics (OTEs),<sup>20–24</sup> lithium-ion batteries (LIBs),<sup>25–27</sup> and solar cells.<sup>28,29</sup> However, the inherently low solubility of CLPs in common organic solvents due to their strong interchain interactions presents significant challenges for processing and manufacturing. This drawback complicates the preparation of

thin films and impedes the use of solution-based processing techniques widely employed in device fabrication.

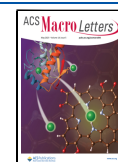
In addition, the insolubility of CLPs, caused by side reactions during their synthesis, remains a significant challenge that limits their development in the field of organic semiconductors. This is why, from 1966 to 2022, only benzimidazobenzophenanthrolinedione ladder (BBL) has been widely applied in OTEs,<sup>20,30</sup> OECTs,<sup>18,31,32</sup> and other semiconductor fields.<sup>33</sup> In 2022, our research group overcame this limitation by utilizing the varying chemical reactivities of functional groups at different temperatures. Through a

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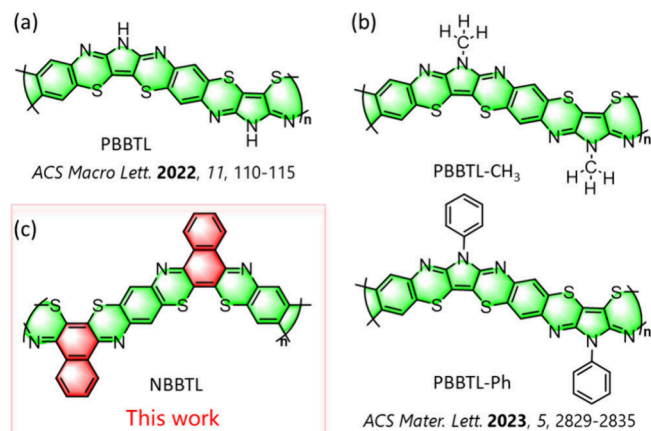
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stepwise polymerization method, we developed a new type of soluble *p*-type fully CLP, 6*H*-pyrrolo[3,2-*b*:4,5-*b'*]bis[1,4]-benzothiazine ladder (PBBTL) (Figure 1a), which was



**Figure 1.** Chemical structures of the reported *p*-type fully CLPs a) PBBTL, b) PBBTL-CH<sub>3</sub>, and PBBTL-Ph, as well as the c) NBBTL discussed in this work.

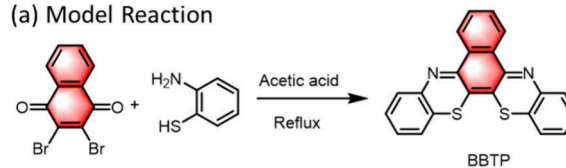
successfully applied to OTEs and OECS.<sup>16,34</sup> Following this, we further modified the pendant groups and synthesized two *p*-type CLPs, PBBTL-CH<sub>3</sub> and PBBTL-Ph (Figure 1b), respectively, whose conductivity and thermoelectric performance were significantly enhanced after doping.<sup>23</sup>

Here, we replaced the pyrrole units in PBBTL with naphthalene units to enhance planarity, resulting in the synthesis of a novel *p*-type ladder-type conjugated polymer, benzo[*a*]benzo[5,6][1,4]thiazino[3,2-*c*]phenothiazine ladder (NBBTL) polymer. The chemical structure of NBBTL was rigorously characterized and validated using chemical model reactions and Fourier-transform infrared (FTIR). The thermoelectric properties of an NBBTL device were evaluated, achieving a maximum electrical conductivity of  $3.18 \pm 0.33 \text{ S cm}^{-1}$  and a power factor of  $0.94 \pm 0.09 \mu\text{W m}^{-1} \text{ K}^{-2}$  after FeCl<sub>3</sub> doping. To gain further insight into the polymer's solid-state packing, two-dimensional grazing-incidence wide-angle X-ray scattering (2D GIWAXS) analysis was conducted. The results revealed a bimodal orientation in the NBBTL thin film, comprising both face-on and edge-on polymer chain packing orientation.

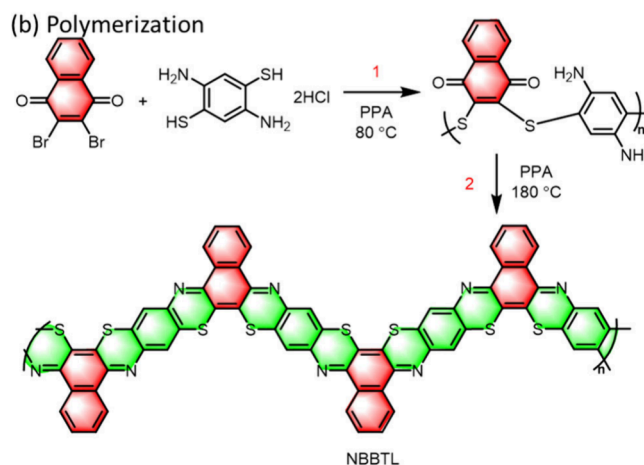
NBBTL was synthesized using similar procedures as PBBTL, PBBTL-CH<sub>3</sub>, and PBBTL-Ph, i.e., reacting equimolar of 2,3-dibromo-1,4-naphthoquinone and the corresponding 2,5-diamino-1,4-benzenedithiol dihydrochloride in mixture solvents of polyphosphoric acid (PPA) and phenylphosphonic acid (PhPO<sub>3</sub>H<sub>2</sub>) (1:1, wt %) (Scheme 1). Due to the poor solubility of NBBTL in conventional solvents such as chloroform, toluene, and trifluoroacetic acid (TFA), precise analysis of its chemical structure using standard techniques like nuclear magnetic resonance (NMR) and high-resolution mass spectrometry (HRMS) is challenging. To overcome this limitation and validate the feasibility of the polymerization reaction, we designed a model reaction under similar conditions. In this reaction (Scheme 1a), 2,3-dibromo-1,4-naphthoquinone was refluxed with two equiv of 2-amino-benzenethiol in acetic acid, yielding the model compound benzo[*a*][1,4]benzothiazino[3,2-*c*]phenothiazine (BBTP) with a high yield of 78%. The high yield of the model reaction also

## Scheme 1. Synthesis of Model Compounds a) BBTP and Ladder-Type Conjugated Polymer b) NBBTL

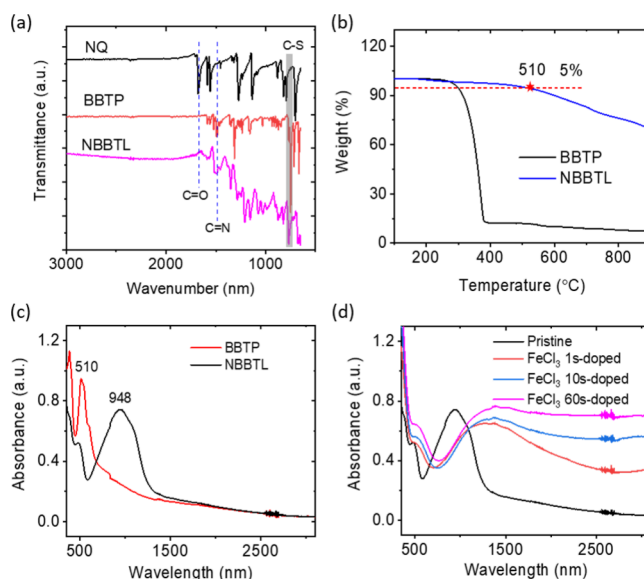
### (a) Model Reaction



### (b) Polymerization



confirms the feasibility of the polymerization reaction. The structure of BBTP was confirmed by <sup>1</sup>H NMR (TFA-*d*<sub>1</sub>) (Figure S1), <sup>1</sup>H NMR (DMSO-*d*<sub>6</sub>) (Figure S2), <sup>13</sup>C NMR (TFA-*d*<sub>1</sub>) (Figure S3), and HRMS (Figure S4). Fourier-transform infrared (FT-IR) was also used to characterize the chemical structures of the starting materials (2,3-dibromo-naphthalene-1,4-dione, NQ), model compounds (BBTP), and polymers (NBBTL), as shown in Figure 2a. The C=O double bond at 1600 cm<sup>-1</sup> in raw material NQ disappeared, while a new C=N and C-S characteristic peak at 1500 cm<sup>-1</sup> and 750



**Figure 2.** (a) ATR-FTIR spectra of NQ (black), BBTP (red), and NBBTL (blue). (b) TGA curve of BBTP and NBBTL under a N<sub>2</sub> atmosphere. (c) UV-vis-NIR absorption spectra of pristine BBTP and NBBTL. (d) UV-vis-NIR spectra of NBBTL films doped with 20 mM FeCl<sub>3</sub> at different times.

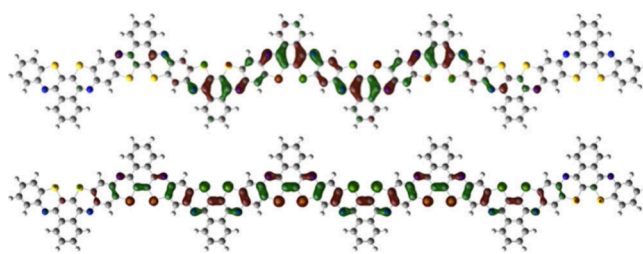
$\text{cm}^{-1}$  appeared in both BBTP and NBBTL, respectively, providing strong support for the conjugated ladder polymer structure (Figure 2a).<sup>21,23,35,36</sup> As NBBTL is only soluble in strong corrosive acids like methanesulfonic acid (MSA), established techniques for molecular weight determination such as gel permeation chromatography and light scattering are not suitable. The intrinsic viscosity has been the recognized parameter as an indirect indicator to reflect for the molecular weight of ladder conjugated polymers via the Mark–Houwink equation.<sup>4,21</sup> The intrinsic viscosity of NBBTL in MSA at 25 °C was determined to be 1.59 dL/g by using an Ubbelohde viscometer.

The thermogravimetric analysis (TGA) curves of BBTP and NBBTL are presented in Figure 2b. BBTP demonstrated significantly lower thermal stability, with a 5% weight loss occurring at 303 °C under a  $\text{N}_2$  atmosphere; the NBBTL exhibited exceptional thermal stability, with a weight loss of less than 5% at 510 °C under a  $\text{N}_2$  atmosphere. In addition, no phase transition was observed below 300 °C in second cycle differential scanning calorimetry (DSC) measurements (Figure S5), indicating the highly rigid structure of NBBTL. This outstanding thermal stability can be attributed to the extensive ladder-type conjugation structure.<sup>34</sup>

The maximum ultraviolet–visible–near-infrared (UV–vis–NIR) absorption ( $\lambda_{\text{max}}$ ) of NBBTL and BBTP thin films were determined to be 948 and 510 nm, respectively (Figure 2c). The  $\lambda_{\text{max}}$  of NBBTL (948 nm) was red-shifted by 438 nm as compared to BBTP, indicating an extension of  $\pi$ -conjugation in polymeric NBBTL as compared to the BBTP small molecule. From their UV–vis–NIR absorption spectra, the optical bandgap ( $E_g$ ) of NBBTL was calculated to be 0.995 eV. To investigate the  $\text{FeCl}_3$  doping process, the UV–vis–NIR absorption spectra of NBBTL films were recorded at various immersion times (Figure 2d). After doping with the  $\text{FeCl}_3$  solution, the neutral-state  $\pi \rightarrow \pi^*$  absorption peak at 950 nm disappeared and a new polaron peak at 1250 nm emerged. This phenomenon can be attributed to the small molecular size of  $\text{FeCl}_3$ , which facilitates its diffusion and penetration into the NBBTL film, enabling efficient doping. As the immersion time increased, the intensity of the polaron peak grew, indicating a higher concentration of polarons in the film.

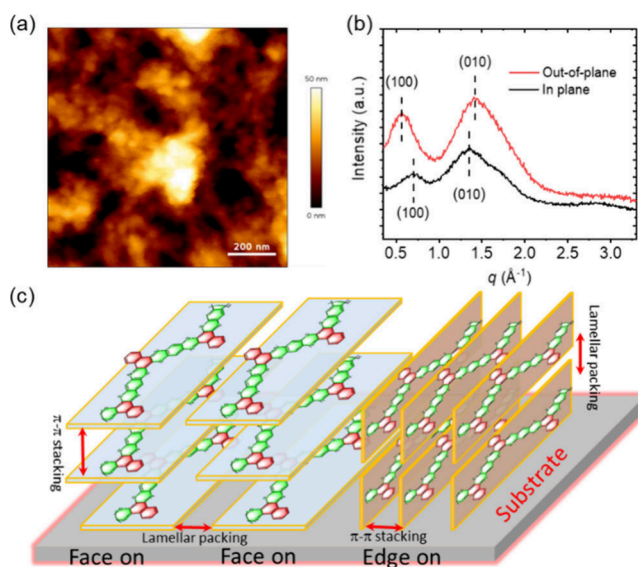
The highest occupied molecular orbital (HOMO) value of NBBTL was evaluated by cyclic voltammetry (CV) curves of NBBTL, BBTP, and ferrocene (Figure S6). The oxidation peak potentials of NBBTL and BBTP, measured with respect to  $\text{Fc}/\text{Fc}^+$  (HOMO =  $-4.80$  eV) as the reference, were determined to be 0.73 and 0.80 V, corresponding to HOMO energy levels of  $-5.53$  eV and  $-5.60$  eV, respectively. To gain deeper insights into the structure of the NBBTL ladder-conjugated polymer, density functional theory (DFT) calculations were performed to optimize and analyze molecular geometry. To simplify the computational workload, a tetramer was utilized instead of the full polymer, and the CAM-B3LYP/6–31G(d,p) basis set was employed. The optimized geometry of the tetramer, shown in Figure 3a, reveals that the NBBTL main chain adopts a planar zigzag configuration. This highly coplanar structure facilitates efficient  $\pi$ - $\pi$  stacking between polymer chains. The calculated HOMO energy level of  $-5.86$  eV closely aligns with the  $-5.53$  eV value obtained from the CV measurements (Figure 3b).

The surface morphology of thin films was characterized by using Atomic Force Microscopy (AFM) techniques. The AFM image demonstrates that the thin film is a porous



**Figure 3.** LUMO and HOMO of the DFT-optimized structure of NBBTL at the CAM-B3LYP6–31G(d,p) level of theory in the gas phase.

nanostructure with dimension of  $\sim 50$  nm, which is consistent with the results of previous experiments.<sup>22,23</sup> The thermovoltage measurements of the NBBTL films doped by  $\text{FeCl}_3$  solution at different immersion times were shown in Figure S7. Additionally, Grazing-Incidence Wide-Angle X-ray Scattering (GIWAXS) measurements were performed on the thin films, as shown in Figure 4b. The presence of distinct dual



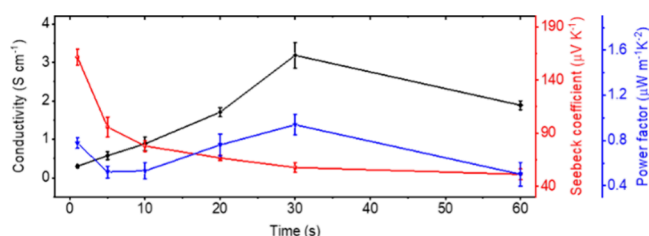
**Figure 4.** (a) AFM height image. (b) GIWAXS characterization of NBBTL films in the out-of-plane and in-plane direction based on Si substrates. (c) Schematic of edge-on and mixture of face-on and edge-on orientation.

lamellar and  $\pi$ - $\pi$  stacking peaks in both the in-plane and out-of-plane directions indicates that the NBBTL molecular chains in the thin film exhibit a bimodal orientation, that is, both face-on and edge-on orientations. The schematic representation of the polymer chain arrangement is shown in Figure 4c. Based on the values of  $q_{xy} = 1.40 \text{ Å}^{-1}$  and  $q_z = 0.567 \text{ Å}^{-1}$ , the intermolecular  $\pi$ - $\pi$  stacking distance was calculated to be 4.488 Å, and the interlayer stacking distance is  $z$ . The  $\pi$ - $\pi$  stacking distance is significantly higher than that of the previously reported  $p$ -type CLPs, PBBTL (3.4 Å). This may be due to the fact that the NH groups of the pyrrole units in PBBTL can interact with each other to form hydrogen bonds. The large  $\pi$ - $\pi$  distance of NBBTL may facilitate the diffusion and enhanced doping efficiency of the smaller-sized  $\text{FeCl}_3$  molecules into the NBBTL molecular chains. However, the penetration of  $\text{FeCl}_3$  may disrupt the ordered structure of NBBTL, leading to overdoping. The phenomenon is evidenced by the disappearance of the neutral-state  $\pi \rightarrow \pi^*$  absorption peak at 950 nm



after just 1 s of doping (Figure 2d), as well as the decline in the thermoelectric performance with increasing doping time (*vide infra*).

Thermoelectric properties of the NBBTL films were measured after *p*-doping in an FeCl<sub>3</sub> solution; the schematic of the Seebeck coefficient measurement is shown in Figure S8. Before doping, the intrinsic conductivity of the NBBTL film was less than 10<sup>−5</sup> S cm<sup>−1</sup>, making it indeterminate for accurate measurement. NBBTL films were doped using a 20 mM FeCl<sub>3</sub> solution in anhydrous acetonitrile, with the doping level controlled by varying the immersion time. After doping, the films were annealed at 80 °C for 10 min to ensure complete evaporation of acetonitrile and stabilize the film, thereby enhancing the reliability of test data. At an immersion time of 1 s, the conductivity of NBBTL increased to 0.30 ± 0.04 S cm<sup>−1</sup>, representing a four-order-of-magnitude enhancement compared to the undoped film.<sup>15,20</sup> The conductivity continued to rise with increasing immersion time, reaching a peak value of 3.18 ± 0.33 S cm<sup>−1</sup> at 30 s. However, further extending the immersion time led to a decrease in conductivity, likely due to overdoping. Besides, the positive Seebeck coefficient of the FeCl<sub>3</sub>-doped NBBTL film confirmed its *p*-type thermoelectric behavior, though the Seebeck coefficient decreased with increasing immersion time, consistent with previous studies.<sup>34</sup> Since the power factor (PF =  $\sigma S^2$ ) depends on both conductivity and the Seebeck coefficient, optimizing the immersion time is critical to maximize thermoelectric performance. As shown in Figure 5, the maximum power factor of 0.94



**Figure 5.** Thermoelectric properties of the NBBTL doped by 20 mM FeCl<sub>3</sub> at different doping time.

± 0.09 μW m<sup>−1</sup> K<sup>−2</sup> was achieved at an immersion time of 30 s. A comparative summary of the film properties between the PBBTL polymer reported in the literature and the NBBTL polymer is presented in Table 1.

In conclusion, we have successfully synthesized a novel soluble conjugated ladder-type polymer, NBBTL. A comparison with model compounds BBTP reveals that NBBTL exhibits minimal structural defects and enhanced thermal stability. To further investigate its microstructure and thermoelectric properties, we spin-coated the polymer onto to a glass substrate and evaluated its thermoelectric performance after FeCl<sub>3</sub> doping. The positive Seebeck coefficient confirmed its *p*-type thermoelectric behavior. The polymer achieved a maximum conductivity of 3.18 ± 0.33 S cm<sup>−1</sup> and a peak

power factor of 0.94 ± 0.09 μW m<sup>−1</sup> K<sup>−2</sup> after a 30 s soaking time. AFM analysis revealed a nanoporous microstructure, while GIWAXS demonstrated a mixed molecular chain orientation in the NBBTL film consisting of both face-on and edge-on structures. The calculated  $\pi$ - $\pi$  stacking distance of 4.488 Å was notably larger than that of PBBTL and BBL (3.4 Å), facilitating the penetration of small FeCl<sub>3</sub> molecules and enhancing doping efficiency. This finding was supported by UV-vis-NIR spectroscopy, which showed the complete disappearance of the neutral  $\pi \rightarrow \pi^*$  absorption peak and the emergence of a new polaron peak at 1250 nm after soaking for just 1 s. We hope that our work will inspire further research and innovative ideas, driving the development of conjugated ladder-type polymers for enhanced performance in thermoelectrics and other applications.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmacrolett.5c00191>.

Materials and syntheses of experiments, characterization data, NMR spectra, HR-MS data, DSC plot, cyclic voltammograms, plotted thermovoltage data, schematic of the Seebeck coefficient measurement set-up (PDF)

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**Table 1.** Comparison of the PBBTL polymer and NBBTL polymer films reported in literature and in this work

| Polymer               | $\lambda_{\max}$ (nm) | HOMO  | $\sigma_{\max}$ (S cm <sup>−1</sup> ) | $E_g$ (eV) | PF (μW m <sup>−1</sup> K <sup>−2</sup> ) | Orientation | Reference                            |
|-----------------------|-----------------------|-------|---------------------------------------|------------|------------------------------------------|-------------|--------------------------------------|
| PBBTL                 | 747                   | −5.47 | 4.50 ± 0.21                           | 1.51       | 7.73                                     | Face-on     | ACS Macro Lett. 2022, 11, 110–115    |
| PBBTL-CH <sub>3</sub> | 725                   | −5.00 | 36.29 ± 0.58                          | 1.46       | 22.7 ± 2.1                               | Edge-on     | ACS Mater. Lett. 2023, 5, 2829–2835. |
| PBBTL-Ph              | 719                   | −4.99 | 1.40 ± 0.16                           | 1.55       | 1.7 ± 0.2                                | bimodal     |                                      |
| NBBTL                 | 948                   | −5.53 | 3.18 ± 0.33                           | 0.995      | 0.94                                     | bimodal     | This Work                            |

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## Author Contributions

Q.H. and J.W. made equal contributions. The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript. CRediT: Jianing Wang formal analysis, writing – original draft; Teck Lip Dexter Tam investigation, methodology, project administration; Jiayu Wang data curation.

## Notes

The authors declare no competing financial interest.

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